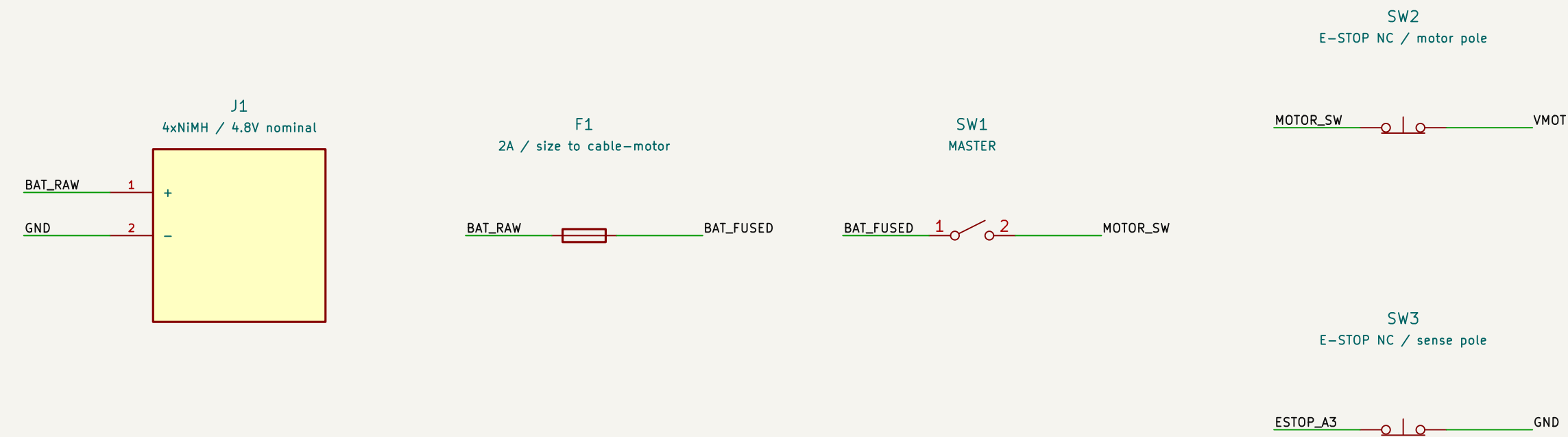
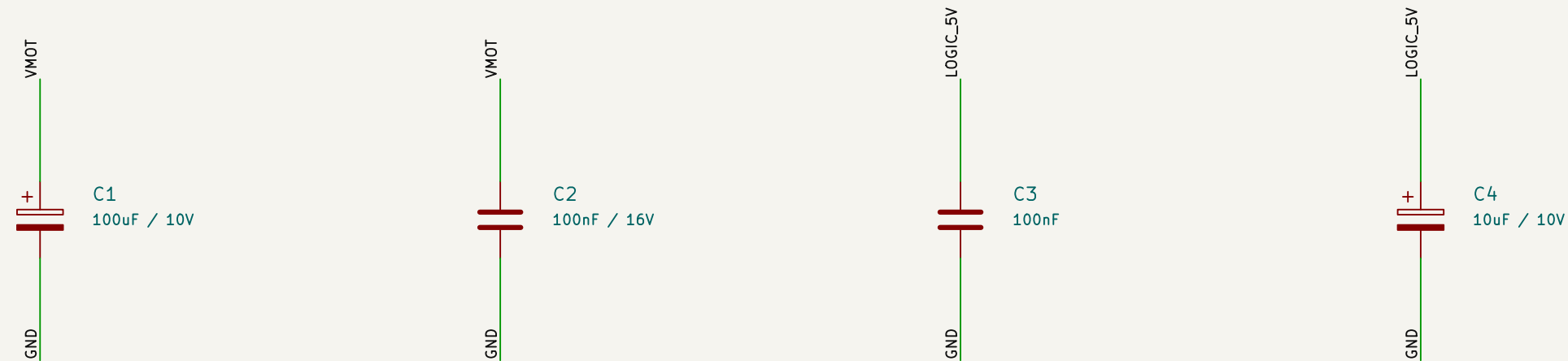


01 POWER / INDEPENDENT MOTOR DISCONNECT

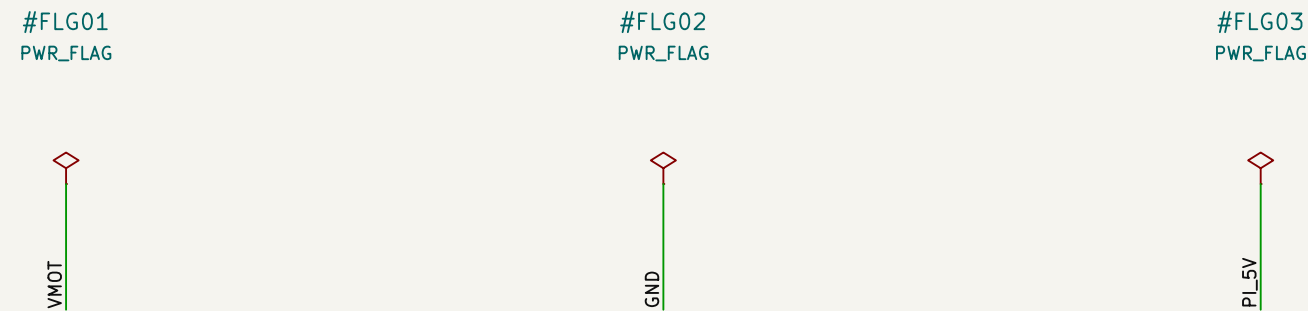


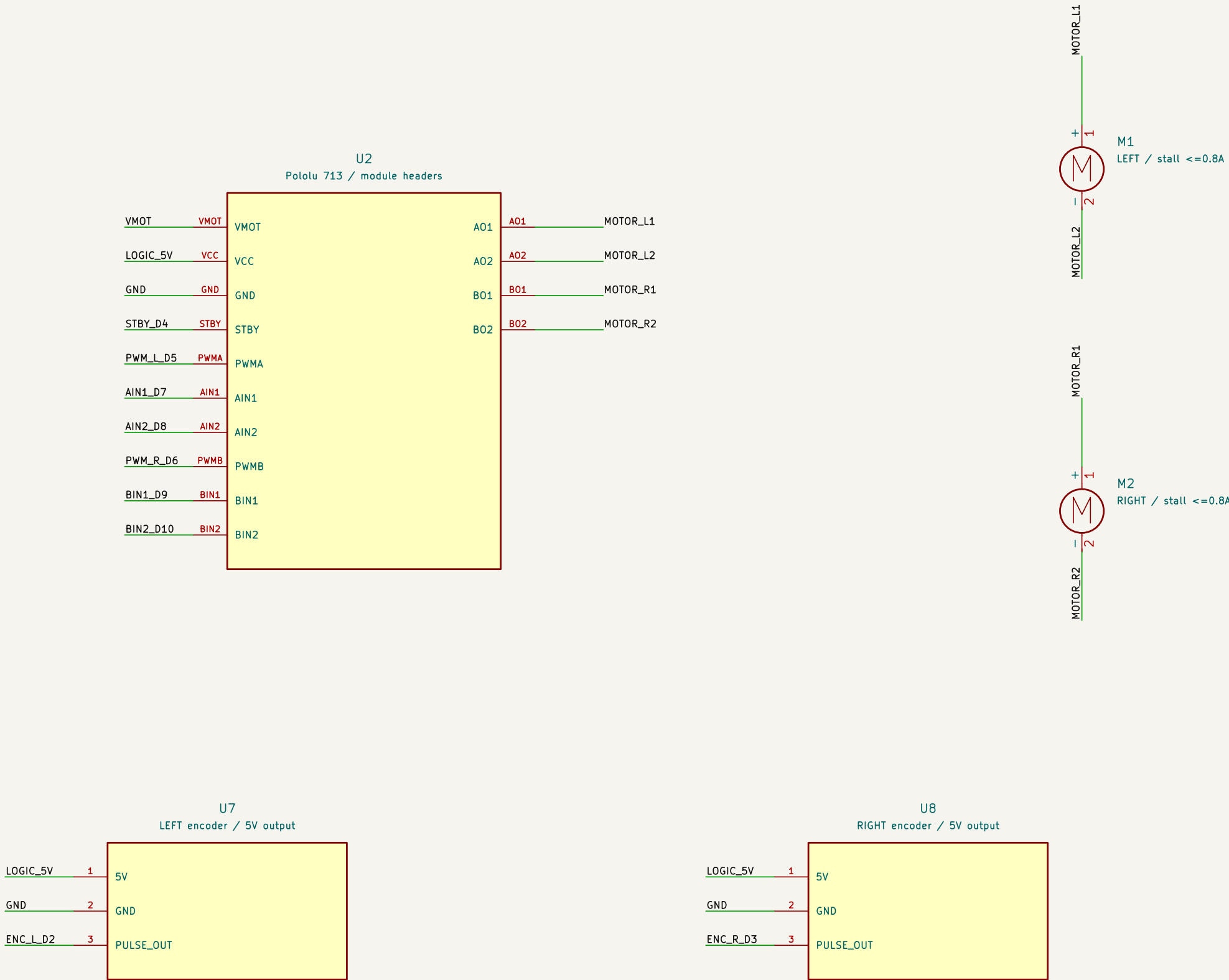
SW2 + SW3 = two isolated NC contacts on ONE latching actuator.

DC motor contact rating >= combined stall current. Reset to rearm.

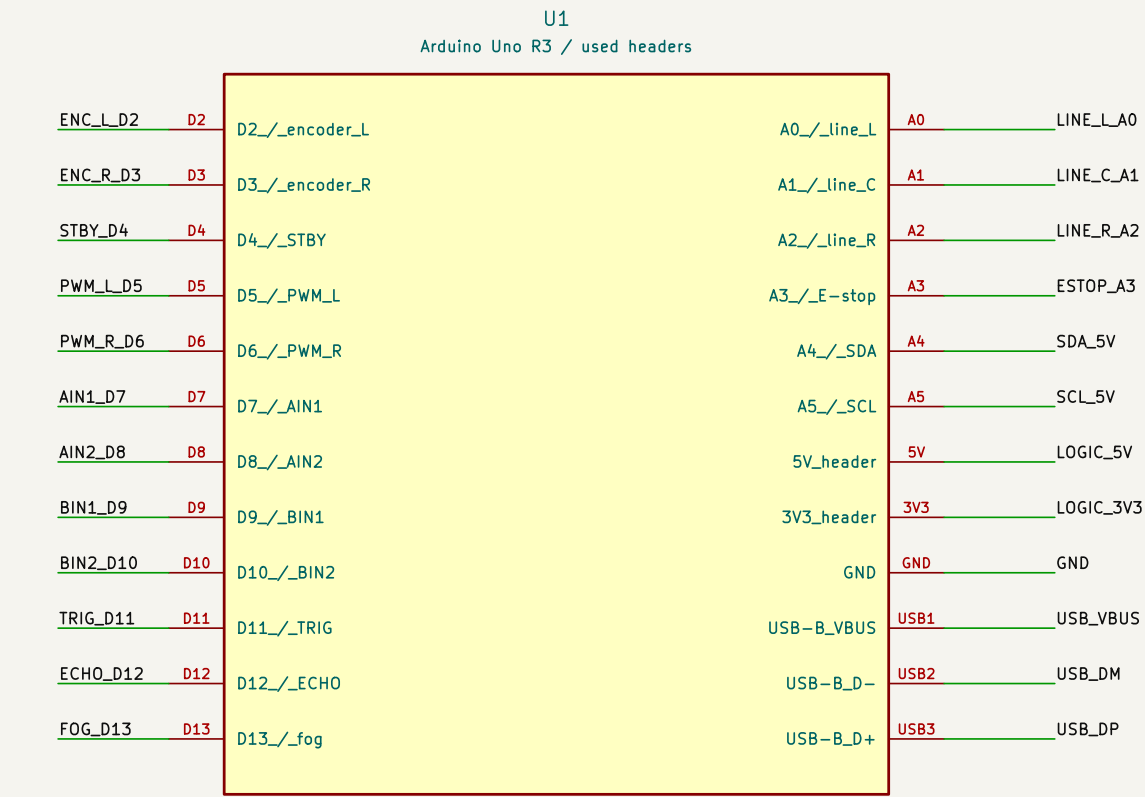


External decoupling; carrier/module onboard capacitors retained.



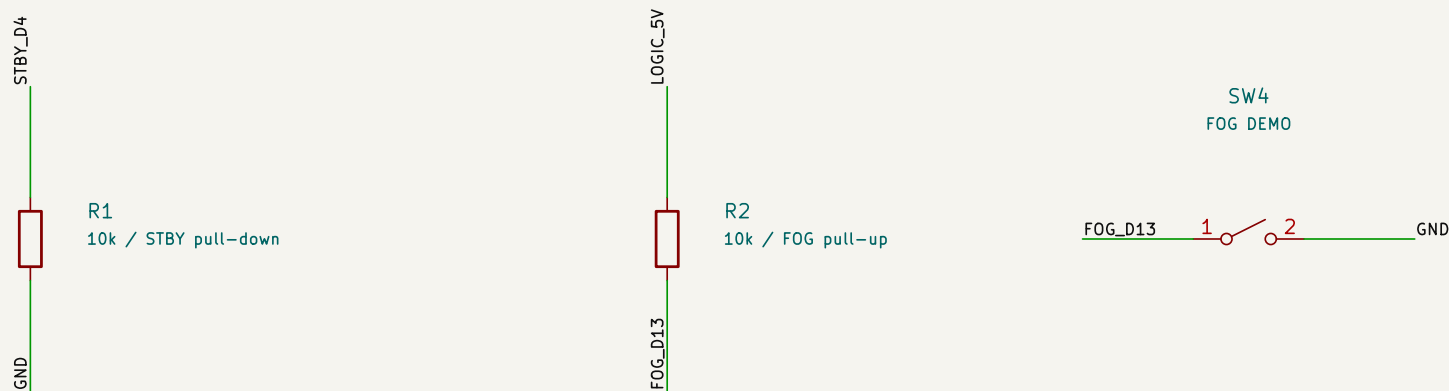


Header module symbols; unused board headers omitted. Not bare-IC pin numbering.

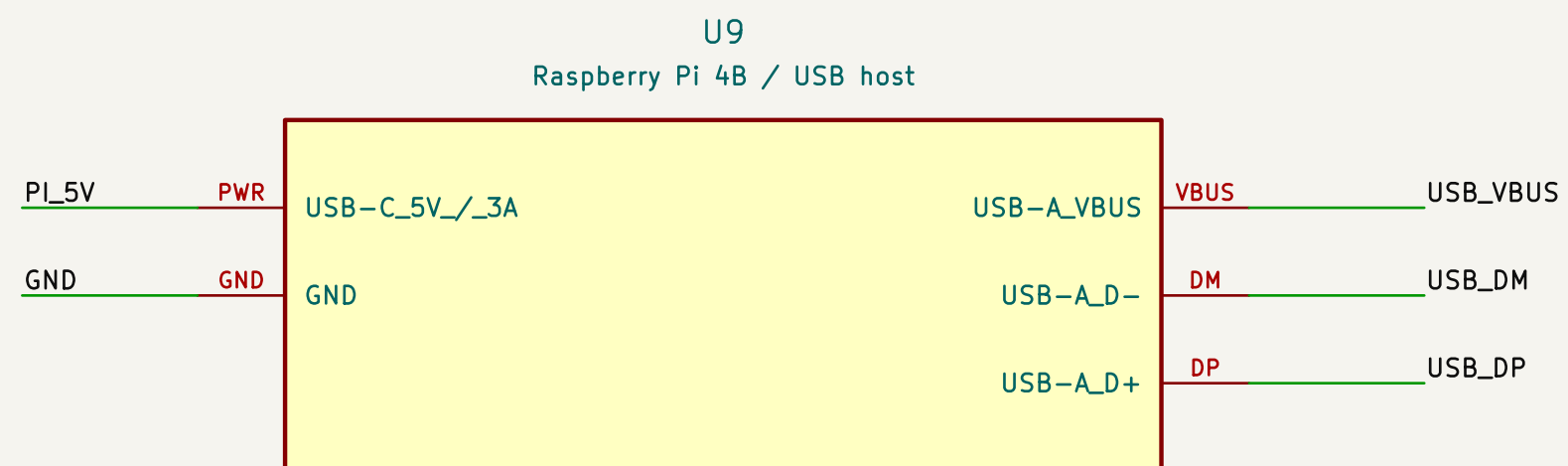
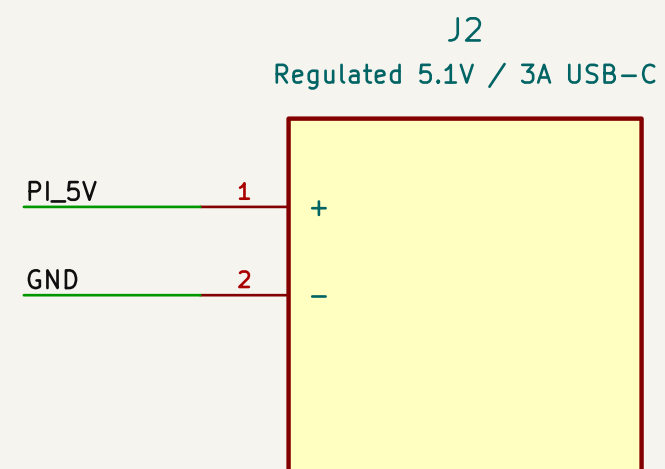


A3, D2, D3: internal pull-ups enabled in firmware.

USB powers Uno. No second 5V feed / no VIN motor feed.



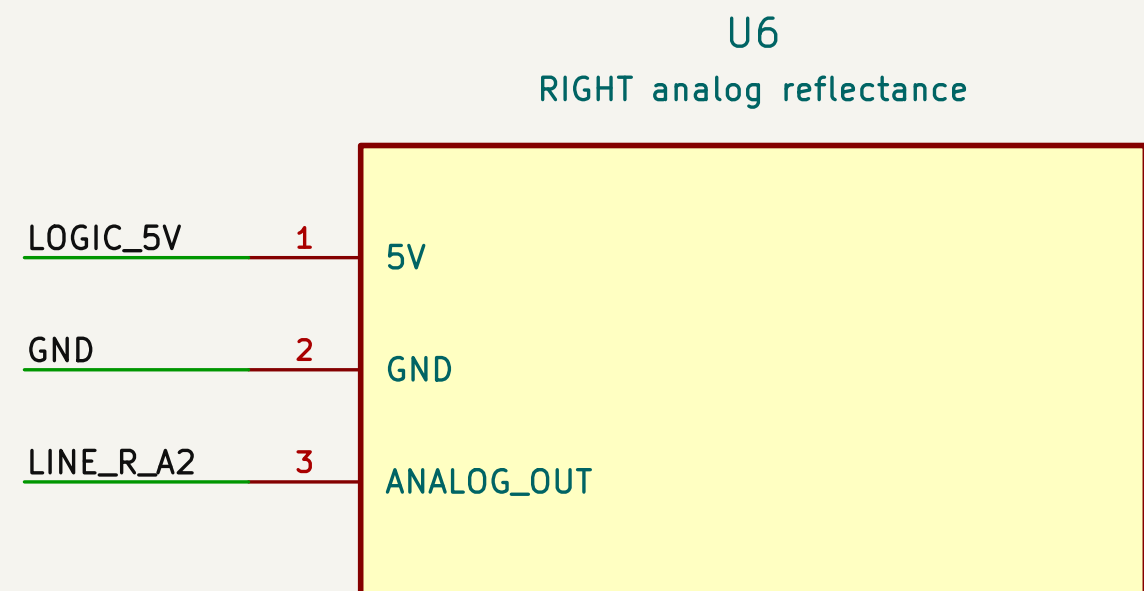
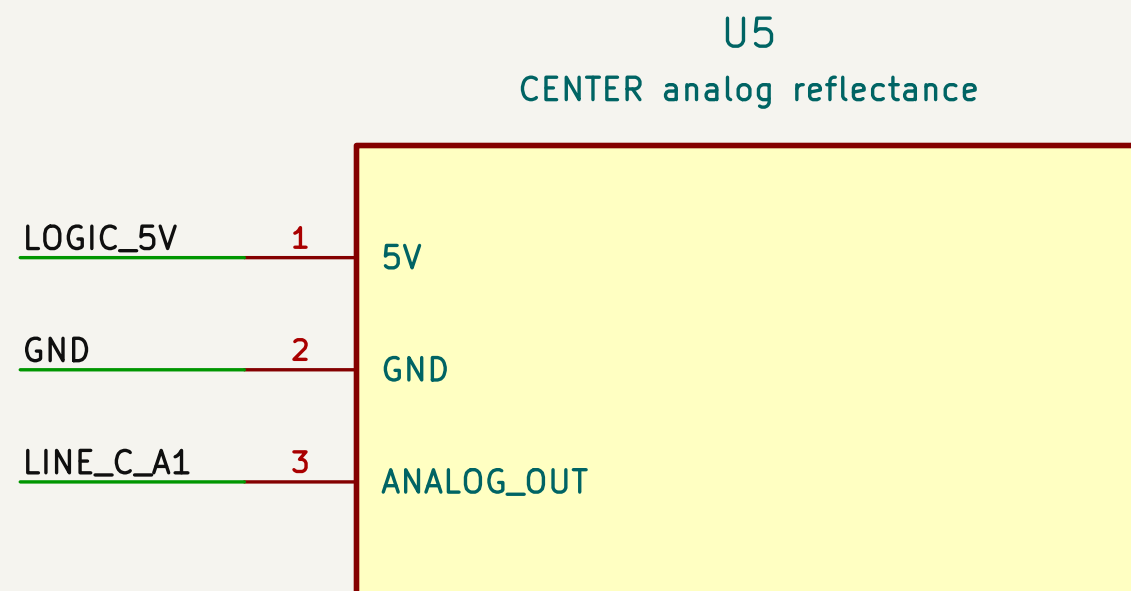
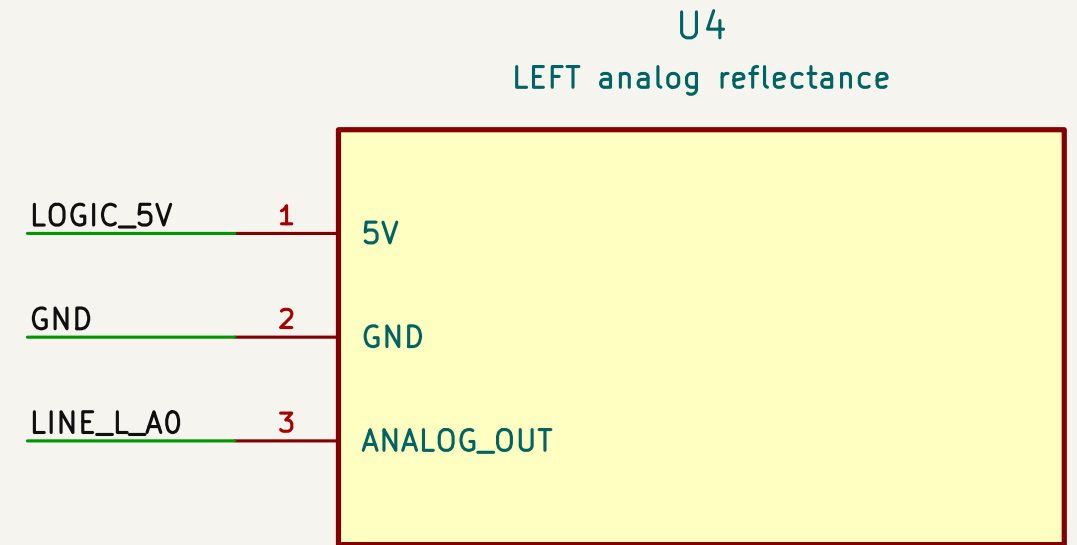
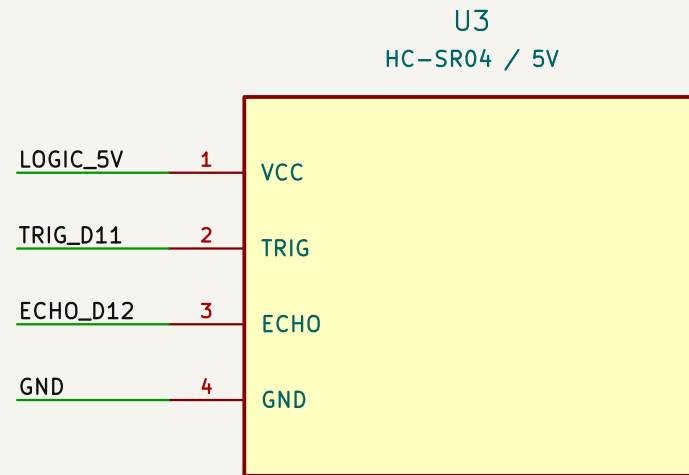
Drive disabled 300ms after last valid USB serial command.



USB-A to USB-B data cable; D+/D- stay inside shielded cable.

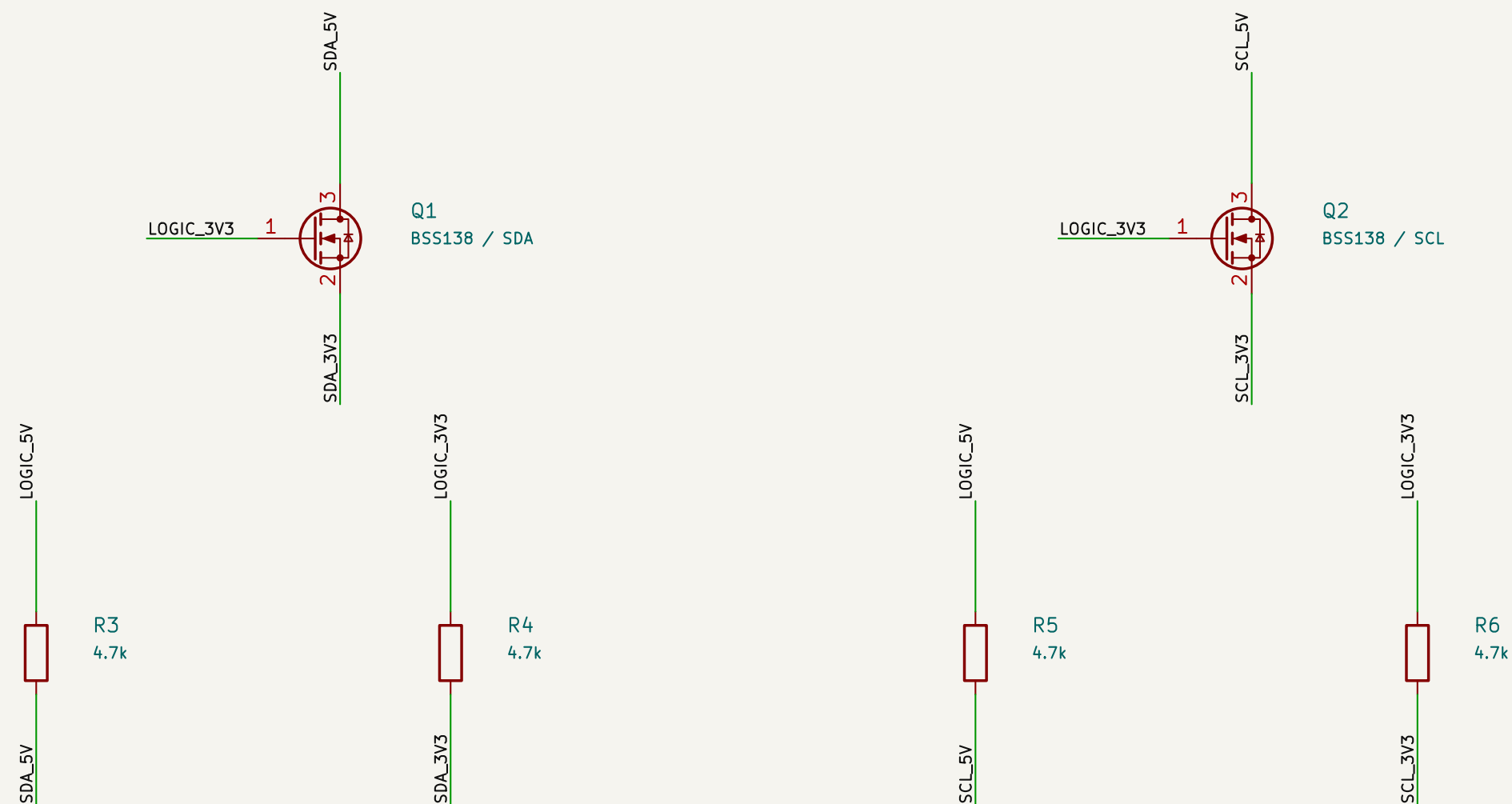
Wi-Fi: V2V UDP 5005 / telemetry UDP 5006.

## 04 RANGE / LINE SENSORS



Analog outputs: 0–5V. HC-SR04 ECHO goes to Uno D12, never Pi GPIO.

05 MPU6050 / DISCRETE I2C LEVEL SHIFT



Q1/Q2: gate=3V3, source=LV, drain=HV; shared ground.

